DOMAIN : HEALTHTECH & DIGITAL HEALTH

MINEARENA 1.0 — PROBLEM STATEMENT

PROBLEM TITLE :

PS-03: Personalized Post-Discharge Deterioration Monitor

EVENT	DURATION	DATES
Code Commit, BIET Davangere	24-Hour On-Site Hackathon	12–13 November 2026

1. PROBLEM OVERVIEW

Patients can experience health deterioration in the days following hospital discharge, when continuous clinical monitoring is no longer available.

Remote patient monitoring can help healthcare professionals identify early warning signs. However, continuously collecting patient vital signs creates another challenge: distinguishing genuine health deterioration from temporary fluctuations, sensor errors and unreliable readings.

Excessive false alerts can create alert fatigue, making it difficult for clinicians to identify patients who genuinely require attention.

The challenge is not simply collecting patient vitals but intelligently identifying when a reading requires clinical attention.

2. PROBLEM STATEMENT

Build an intelligent post-discharge patient monitoring system that continuously receives wearable vital-sign readings, analyses patient-specific health conditions and medications, detects early signs of deterioration and generates meaningful alerts for healthcare professionals.

The system must ingest patient readings as FHIR R4 Vital Signs Observations, including:

- Heart Rate (LOINC: 8867-4)
- Oxygen Saturation – SpO2 (LOINC: 2708-6)
- Respiratory Rate (LOINC: 9279-1)
- Body Temperature (LOINC: 8310-5)

No physical hardware is required.

The system must support two software-based data sources:

1. Replayed recordings from the open BIDMC PhysioNet dataset.
2. A virtual wearable simulator capable of streaming patient readings and injecting different sensor events.

Each patient must have a FHIR discharge profile containing medical conditions and medications, generated using Synthea.

The monitoring engine must calculate a NEWS2-style early-warning score as its baseline and personalize the scoring using patient-specific health information.

For example, patients with documented hypercapnic respiratory failure must use SpO2 Scale 2 with a target range of 88–92%.

The system must also maintain individual patient rolling baselines to identify gradual deterioration rather than relying only on sudden changes in individual readings.

Alerts must be categorized into three levels:

- Watch
- Review
- Urgent

Every alert must explain the reading, the rule applied and the patient profile information that triggered it.

The system must distinguish genuine deterioration from sensor artifacts, suppress false alerts and provide a clinician dashboard for monitoring patients.

Important: This system is intended for clinical decision support, not medical diagnosis. This limitation must be clearly displayed in the user interface.

3. MUST-HAVE REQUIREMENTS (MANDATORY)

All the following requirements must be implemented and demonstrated.

3.1 FHIR R4 Ingestion API

- Build an API that accepts patient vital-sign readings as FHIR R4 Vital Signs Observations.
- Support heart rate, oxygen saturation, respiratory rate and body temperature.
- Store the received observations in a HAPI FHIR server.
- Ensure that the stored observations follow the required FHIR structure and LOINC coding.

3.2 Virtual Wearable Simulator and BIDMC Replay

Implement two software-based data ingestion sources.

A. Virtual Wearable Simulator

Build a simulator that streams readings for individual patients at a configurable rate.

The simulator must support scripted events such as:

- Gradual vital-sign drift.
- Sudden spikes.
- Missing readings or dropouts.
- Motion-related noise.
- Probe removal artifacts.

For probe removal, simulate SpO2 as zero or null while the heart rate flatlines.

Body temperature readings must be generated using the simulator because the BIDMC dataset does not provide body temperature.

B. BIDMC Dataset Replay

- Integrate the open BIDMC PhysioNet dataset.
- Replay recorded heart rate, respiratory rate and SpO2 readings.
- Support real-time or accelerated playback.
- Stream the readings into the FHIR ingestion API.

3.3 Patient Discharge Profiles

- Generate at least 5 synthetic patient profiles using Synthea.
- Each profile must contain relevant medical conditions and medications.
- Patient profiles must influence how the monitoring system calculates scores and generates alerts.

3.4 NEWS2-Based Scoring Engine

- Implement a NEWS2-style early-warning scoring baseline.
- Use patient-specific information to personalize the scoring.
- Implement at least two personalization rules.

Examples:

1. Apply SpO2 Scale 2 (target 88–92%) for patients with documented hypercapnic respiratory failure.
2. Maintain individual patient rolling baselines to identify gradual deterioration instead of relying only on sudden vital-sign spikes.

3.5 Sensor Artifact Detection and Handling

- Detect implausible vital-sign readings.
- Identify flatlined signals and sensor-related artifacts.
- Detect probe removal scenarios.
- Suppress false clinical alerts caused by unreliable sensor readings.
- Distinguish sensor-related issues from genuine patient deterioration.

3.6 Clinician Dashboard and Alert Management

Build a working clinician dashboard that provides:

- Patient monitoring and vital-sign visualization.
- Patient-specific health information.
- NEWS2-style scoring results.
- Tiered alerts: Watch, Review and Urgent.
- Clear explanations for every generated alert.
- The reading, rule and patient profile fact responsible for each alert.

3.7 Scenario Replay and Performance Report

- Support replay of scripted scenarios.
- Include stable recovery, slow deterioration and sensor artifact scenarios.
- Generate a performance report comparing the proposed system against naive fixed-threshold monitoring.
- Report detection lead time and alerts per patient-day.

4. EXPECTED SYSTEM WORKFLOW

The system should demonstrate the following end-to-end process:

STEP 1	**Patient Profile Generation:** Generate at least 5 synthetic patient discharge profiles using Synthea.
STEP 2	**Vital-Sign Data Generation:** Receive readings from the virtual wearable simulator or replay the BIDMC PhysioNet dataset.
STEP 3	**FHIR Ingestion:** Convert and submit patient vital readings as FHIR R4 Vital Signs Observations through the ingestion API.
STEP 4	**Data Storage:** Store the observations in the HAPI FHIR server.
STEP 5	**Health Scoring:** Calculate the NEWS2-style baseline score and personalize it using patient-specific conditions and medications.

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STEP 6	**Deterioration Detection:** Analyse vital-sign trends and individual rolling baselines to identify early signs of deterioration.
STEP 7	**Artifact Detection:** Identify sensor errors, flatlined signals, missing readings and motion-related noise.
STEP 8	**Alert Generation:** Generate Watch, Review or Urgent alerts with clear explanations.
STEP 9	**Clinician Dashboard:** Display patient information, vital trends, scores and explained alerts.
STEP 10	**Performance Evaluation:** Replay test scenarios and compare detection lead time, false alerts and alerts per patient-day against fixed-threshold monitoring.

★ HINT – You can implement the following features as well.

Patient Web Application

- Build a patient-facing web application.
- Allow patients to complete daily symptom check-ins.
- Include symptoms such as breathlessness, fever and dizziness.
- Integrate symptom responses into the monitoring score.

Automated Clinical Team Escalation

- Implement escalation to the healthcare team.
- Use FHIR Flag or CommunicationRequest resources.
- Support Kannada or Hindi notifications.

Missing-Data Alerts

- Detect when a patient's wearable device stops sending readings for an extended period.
- Generate separate alerts for missing device data.
- Clearly distinguish device connectivity issues from clinical deterioration alerts.

5. EVALUATION CRITERIA

Judges will evaluate the solution based on the following parameters:

1	**Detection Lead Time:** How early the system identifies slow-deterioration scenarios before they become more severe.
2	**Alert Efficiency:** Number of alerts generated per patient-day compared with naive fixed-threshold monitoring. Lower is better when detection performance is maintained.
3	**False Alert Detection:** Ability to identify and suppress false alerts caused by sensor artifacts, probe removal and unreliable readings.
4	**FHIR Compliance:** Correctness and validity of stored FHIR R4 Vital Signs Observations.
5	**Alert Explainability:** Clarity of every alert, including the triggering vital reading, applied rule and relevant patient profile information.

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6. SUGGESTED TECHNOLOGY STACK

The following technologies are recommended but not mandatory.

FHIR Server	HAPI FHIR R4.	**Patient Data Generation**	Synthea.
Programming Language	Python.	**Backend**	FastAPI.
Real-Time Streaming	WebSockets.	**Dataset**	BIDMC PPG and Respiration Dataset.
Virtual Wearable	Custom Python Simulator.	**Dashboard**	Streamlit or React.
Database	HAPI FHIR server storage.		

All suggested components are free and open source.

Teams are free to choose equivalent technologies if all mandatory requirements are satisfied.

7. DATA SOURCES AND REFERENCES

1. **HL7 FHIR R4 – Vital Signs Profile:** https://hl7.org/fhir/R4/observation-vitalsigns.html
2. **Royal College of Physicians – NEWS2 Chart:** https://www.rcp.ac.uk/file/9436/download
3. **PhysioNet – BIDMC PPG and Respiration Dataset:** https://www.physionet.org/content/bidmc/1.0.0/
4. **2025 Meta-Analysis – Remote Biometric Sensing and Readmission After Discharge:** https://doi.org/10.1002/jhm.70224
5. **JAMA Network Open – Remote Monitoring After Infection and Sepsis, 2026 Trial:** https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2850152
6. **Synthea – Synthetic Patient Generator:** https://synthetichealth.github.io/synthea/

8. IMPORTANT INSTRUCTIONS

- All code must be written during the 24-hour hackathon.
- No physical wearable hardware is required.
- Open-source libraries, public datasets and APIs are allowed.
- Teams must implement both the virtual wearable simulator and BIDMC dataset replay tool.
- All mandatory requirements must be demonstrated during evaluation.
- Suggested technologies are not compulsory.
- The system must clearly state that it provides clinical decision support and is not a medical diagnosis system.
- Teams must demonstrate working functionality rather than only slides, mock-ups or a presentation.

FINAL EXPECTATION

Build an intelligent, personalized post-discharge monitoring system that can continuously process patient vital signs, understand individual patient health profiles, detect gradual deterioration, distinguish sensor artifacts from genuine health concerns and generate meaningful, explainable alerts for clinicians.

Monitor continuously. Detect deterioration early.
Reduce false alerts. Prioritize patient safety.